

Differences therefore arise from how each SOC branch consumes the shared health context. DM is the low-complexity integration reference. HDM isolates the effect of learned capacity adaptation without introducing voltage feedback. HECM adds model-based voltage-residual correction to current-driven state propagation. DD represents compact temporal learning. The GRU is used because SOC estimation and disturbance recovery depend on recent signal history while its gated single-state structure remains lighter than a comparable LSTM. Since DD also consumes the engineered cumulative-charge feature Q_c , it is a data-driven sequence estimator with physics-informed inputs rather than a raw-signal-only neural model.

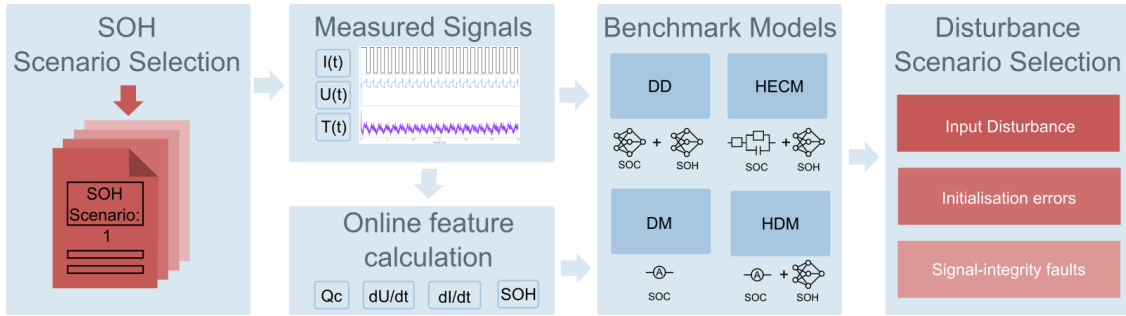


Figure 6.1: Methodology overview of the causal benchmark chain. Measured battery signals are converted into online auxiliary features, processed by the four estimator representatives, exposed to measurement and initialization interventions, and evaluated through global and local performance metrics.

6.2.2 SOC estimation methods

Direct-measurement model (DM)

The direct-measurement formulation used in DM, HDM, and as an explicit auxiliary input to DD follows the same causal cumulative-charge state Q_c . It is updated from the measured current stream as

$$Q_c(k) = Q_c(k-1) + \frac{I_k \Delta t_k}{3600 \text{ s h}^{-1}}, \quad (6.1)$$

where I_k is given in amperes and Δt_k in seconds. The conversion factor produces an ampere-hour increment. The sign convention is fixed by the benchmark implementation, and the state is reset to $Q_c = 0$ after a sustained constant-voltage full-charge event. DM maps this charge state to SOC through the fixed nominal